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1. Basic Features

The energy storage converter has several states: "Initialization," "Self-check," "Shutdown," "Soft Start," "Operation," "Fault," and "Test." When the converter transitions between different modes, the state switching principle shown in the following diagram must be satisfied.

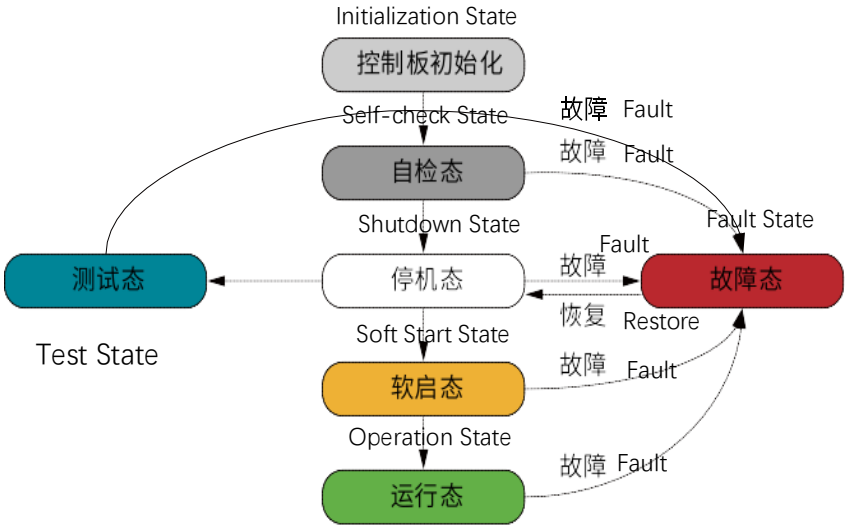


Figure 1-1 Schematic diagram of state transition principle

a) When the converter is powered up and enters initial shutdown, the control system will perform a self-check to verify the integrity of the control and sensor systems.
b) The MCU starts up normally, and the converter enters the key shutdown state. When the power button is in the shutdown state, the energy storage inverter blocks the drive and opens the AC and DC switches.
c) When the host computer sends a power-on command, the energy storage converter enters the soft-start state and performs the AC/DC relay closing and soft-start power-on procedure.
d) After successful soft start, the energy storage converter enters the operational state.

EMS can issue start, standby, and stop commands to PCS on the operation interface, enabling PCS to enter the running state, soft start state, and stop state respectively. After entering the fault state, issuing a reset command through EMS can restore PCS to the stop state.

1.1 Converter Status Description

● **Initialization**: After normal AC/DC connections are established, the converter performs an initialization state upon first power-up, without any command operations

or scheduling. It conducts a current zero-drift self-check and grid status detection. After completing these checks, the system automatically transitions to the self-check state.

● **Self-check:** After normal AC/DC connections are made, the inverter performs a self-check upon first power-up, even without any command or scheduling. Based on the self-check command word, the system enters other sub-states, sequentially executing GFCI self-check, IGBT module Block self-check, insulation resistance detection, and self-check of cooling components (such as fans and pumps). This self-check is performed only once after power-up. If there is no self-check and the naming enable is not set, wait for a period of time before entering the stop state.

● **Shutdown**: After normal AC/DC connection, the converter remains in a button-off state following initial shutdown, with the AC and DC circuit breakers open, even without any command or scheduling. When there are no faults or alarms, the converter accepts local or remote commands and scheduling in shutdown mode. When operational conditions are met, the converter transitions from button shutdown to command-based operation mode. After running, if a shutdown command is received, the converter transitions from running to key shutdown state.

● **Soft Start Standby**: When the converter receives a command or scheduling instruction, the system transitions from the shutdown state to the soft start standby state. Depending on the operating mode and status, DC/AC relay voltage difference checks and relay sticking detection are performed. Once the soft start standby mode is completed, the converter's AC and DC circuit breakers remain closed, but the IGBT does not generate pulses.

● **Operation**: After normal AC/DC connection, the converter enters the operating state following a successful soft start, and performs corresponding work instructions according to commands and scheduling from the upper computer. When a shutdown-type fault occurs, the energy storage converter will open the AC and DC circuit breakers, report an alarm message on the monitoring system, and enter a fault state.

● **Fault:** When the energy storage system fails, the inverter stops operating, immediately opening the AC and DC side switches to disconnect the main circuit of the unit from the battery, the power grid, and the load. The system continuously monitors whether the fault has been cleared. If the fault remains, it stays in the fault state; if the fault is cleared, it automatically recovers after a delay. Under the condition that manual recovery is enabled, after a fault occurs during operation, the fault status can only be cleared by manual reset and pressing the power-off button.

● **Test** After normal AC/DC connection, the converter remains in the button-off state after initial shutdown, with no command operations or scheduling. When there are no faults or alarms, the converter accepts command operations and scheduling from the interface or host computer in shutdown mode. When the host computer triggers a test, the converter transitions from the shutdown state into the command test mode.

1.2 Grid-connected operation function

The energy storage inverter has comprehensive grid-connected operation capabilities and can flexibly adapt to charging and discharging control of various battery types. The charging/discharging modes and commands are set by the host computer or user interface.

● **Charging mode: The PCS** draws AC power from the grid and converts it into DC power to charge the battery. At this time, the energy storage system is connected to the grid, and the direction of energy flow is: grid → PCS → battery.

● **Discharge Mode**: When the grid is operating normally, the PCS converts the DC power from the energy storage battery into AC power synchronized with the grid, injecting it into the grid or supplying power to local loads.

Grid-connected mode is set according to dispatch instructions and can operate in either charging or discharging states. A negative power command indicates charging mode, while a positive power command indicates discharging mode.

When the EMS issues power commands through the operating interface, the PCS will enter the discharging or charging mode based on the positive or negative value of the received power commands.

2. EPO function

The energy storage inverter is equipped with an emergency stop function. If an unexpected situation arises requiring the immediate shutdown of the energy storage inverter, the emergency stop function can be used to immediately halt the inverter's operation.

The emergency stop function provided by the energy storage converter at the factory is a red button on the cabinet door. When the red button on the cabinet door is pressed, the PCS will immediately stop outputting waves, open the AC and DC circuit breakers, and report an emergency stop fault. Subsequently, the state machine transitions to the fault state.

3. CCCV mode

Under this operating mode, the energy storage converter has the capability to regulate charging and discharging currents. It charges or discharges according to a set

current value, with a range of –In to +In. Negative values indicate charging, while positive values indicate discharging. The current value can be set via local or remote commands.

The EMS sends a current command through the operating interface, and after receiving the current command, the PCS will charge or discharge according to the command.

4. PQ mode

Under this operating mode, the energy storage converter has the capability to regulate active power charging and discharging. It charges or discharges according to a set active power command (e.g., -100 kW or +50 kW), and the power value can be set locally or remotely.

EMS issues active power commands through the operating interface, and after receiving the active power commands, PCS will charge or discharge according to the commands.

5. Constant Reactive Power Mode

Under this mode, the energy storage converter continuously injects or absorbs reactive power into the grid according to the set reactive power value (e.g., -100 kVar or +50 kVar).

The EMS issues reactive power commands through the operating interface, and after the PCS receives the reactive power commands, it will inject or absorb reactive power into or from the grid according to the commands.

6. Auto/Manual and Remote Operation function

Energy storage converters support two control modes: local control and remote control. Users can flexibly switch control modes according to on-site requirements.

- **Local Control**: When the energy storage inverter is set to local control mode, it responds only to control commands sent from the HMI (start-up, shut-down, active/reactive power setpoints, etc.).

● **Remote Control**: When the energy storage inverter is set to remote control mode, it only responds to control commands sent from the EMS (such as startup, shutdown, active/reactive power setpoints).

7. Constant PF mode

Under this mode, the energy storage inverter dynamically calculates the required reactive power (Q) based on the real-time active power (P) according to a set power factor (PF = 0.95 or PF = -0.95), always maintaining the power factor at the inverter's output side (PF = P/S) at the set value.

The EMS issues a power factor command through the operating interface, and after receiving the power factor command, the PCS will output power at the set power factor according to the command.

8. Volt-VAr mode

Q_U curve refers to the voltage/reactive power control function, representing the energy storage inverter's capability to provide reactive power support to the grid under voltage fluctuations.

As shown in Figure 2, the Q_U curve achieves voltage/reactive power control functionality by setting different parameters at four inflection points with coordinates (V1, Q1), (V2, Q2), (V3, Q3), and (V4, Q4). Our company's default curve upon factory shipment is (0.92, 0.33), (0.94, 0), (1.06, 0), (1.08, -0.33):

When $U \leq 0.92Un$, $Q = 0.33Pn$;

When $0.94Un \leq U \leq 1.06Un$, $Q = 0$;

When $U \geq 1.08Un$, $Q = -0.33Pn$;

Under other voltage conditions, the reactive power output can vary linearly according to the set curve.

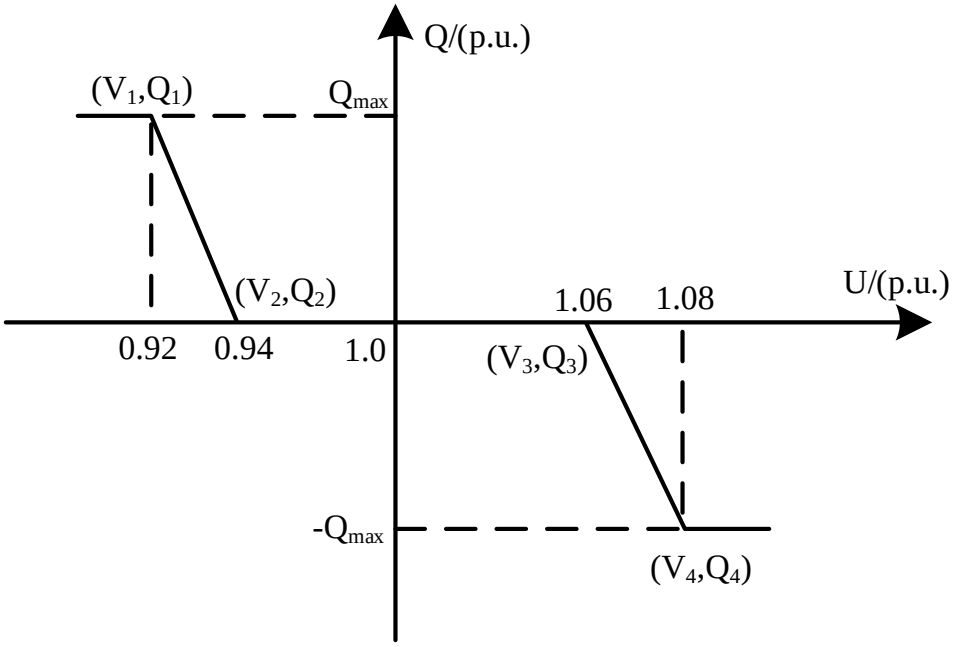


Figure 2 Voltage / Reactive Power Control Curve

9. PF-Watt mode

The PF-P curve is the power factor/active power control function. As shown in the PF-P curve in Figure 3, the default curve provided by our company uses the four inflection points coordinates: (P1, PF1), (P2, PF2), (P3, PF3), (P4, PF4), with the default values being (0.1, 1.0), (0.5, 1.0), (1.0, -0.9), and (1.1, -0.9).

Output: Power factor PF, reactive power reference value Qref

Using the current real-time active power Pg_pu as input, with PF_P_Mode_Enable enabling the PF-P curve, the PF value is output according to the default curve values set in the software, and the reactive power command value is then calculated.

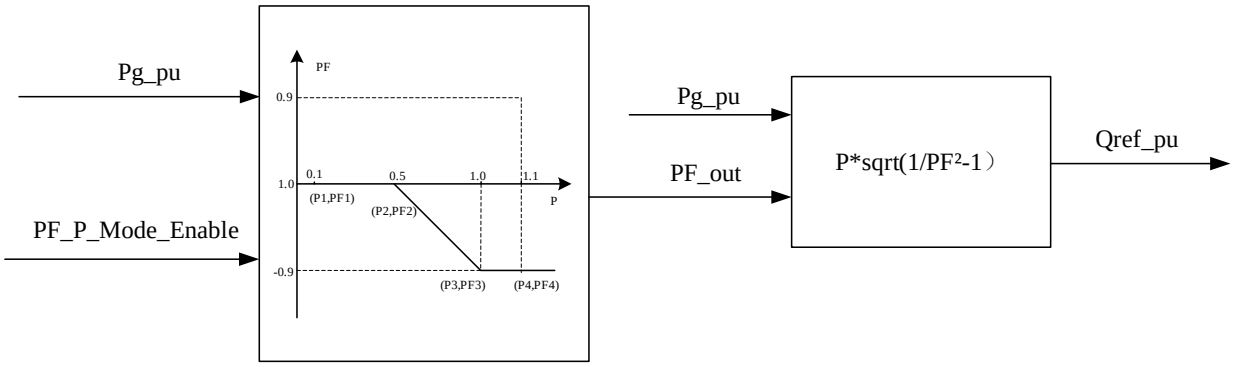


Figure 3 PF/P control curves

10. Black start

10.1 Black Start Introduction

As research and practice in electricity markets deepen, the importance of auxiliary services in power systems for system security is attracting increasing attention. The most severe fault that can occur when an electric power system is operating is a complete system blackout. Once the entire power system is shut down, the final step in power system security measures is how to quickly restore power supply to users. Several major power outages that have occurred worldwide in recent years remind us that studying the recovery of power systems after accidents—namely, black start services—has significant theoretical and practical importance.

Black start refers to the process whereby, after a total system failure causes a complete power outage (except possibly for isolated small grids still operating), the entire system is in a "black" state. Without relying on external network support, the system is gradually restored by starting up power generation units with self-starting capability, which then bring online units lacking self-starting ability, progressively expanding the recovery scope until the entire system is fully restored.

10.2 Black start methods

Depending on the black start service capabilities of different power grids, black start services can be categorized into three types:

(1) The power grid has sufficient black-start capability to autonomously restore power supply according to a black-start recovery plan during a large-scale blackout. This is the common approach used in most systems.

(2) There are no units within the power grid capable of meeting the black-start service requirements, or the number of black-start service units is insufficient to meet the needs of system restoration, but within this geographical region, there are qualified units that do not belong to this power grid. This power grid must sign contracts with black-start service units and the operators of the grids to which the units belong, specifying the methods for providing black-start services and the calculation of fees.

(3) There are no units within the power grid capable of meeting the black-start service requirements, or the number of black-start service units is insufficient to meet the needs of system restoration, and no black-start service units from other power grids

are available. This form of recovery process is relatively passive and can only gradually restore power supply when other grid areas have partially recovered and the system has the capability to provide emergency power support for black-start services to this area. This approach eliminates the cost of maintaining black-start capability, requiring only payment for emergency power support during black-start services. The trade-off is reduced cost, but with extended and uncertain power restoration times. This type of system recovery plan must include a service contract with adjacent power grid areas, specifying the timing and corresponding costs for emergency power support during black start services.

10.3 Black Start Plan

10.3.1 Conditions for black start capability

(1) The auxiliary black-start test of the energy storage system should be fully coordinated with power grid dispatching. Based on each test phase, test scope, and initial conditions (partial AC power loss, complete AC power loss), a test implementation plan should be developed. The plan must clearly define the test power outage scope, safety isolation points, safety measures, and emergency response plans for faults.

(2) The key equipment and systems required for black start are: 1) self-starting power supply; 2) transmission and distribution equipment; 3) monitoring and control systems; 4) communication and coordination equipment; 5) energy storage equipment; 6) auxiliary equipment; 7) critical load facilities. Energy storage system auxiliary black start tests should be conducted according to the power grid black start plan, taking into account the actual conditions of the power station equipment.

(3) The energy storage system (battery, PCS, step-up transformer) equipment is operating normally, with the battery power meeting the maximum power requirement for black start and the battery charge level sufficient to support two black starts.

(4) The diesel generator equipment is operating normally and is capable of starting.

(5) The black start fast switching system is operating normally.

(6) The generator excitation system is operating normally, and all system relay protection devices are in service.

(7) The thermal control DCS system is operating normally, and the black start logic verification is qualified.

(8) During the test period, the unit should be electrically disconnected from the power grid at the primary level, meaning the high-voltage side circuit breaker of the

unit's transformer unit should be in the open position, with the unit operating in an off-grid state.

(9) The control DC system is operating normally.

(10) Dynamic tests should be conducted in the order listed in this standard.

10.3.2 Isolation Scope for Black Start

The isolation scope for black start must comprehensively consider power supply capacity, load demand, power grid structure, and protection configuration, with the core objective of rapidly restoring a stable, independent power island within the smallest possible safe area. In practical applications, safety and reliability of the recovery process must be ensured through simulation rehearsal and dynamic adjustment.

10.3.3 Auto-startup Process

· Preparation phase

Power supply confirmation: Select a power supply with auto-restart capability,

Energy storage power station: self-starts via DC busbar.

Network isolation: Disconnect the tie lines from the main grid to form an initial island.

Communication restored: Activate the backup communication system to ensure dispatch instructions are delivered.

· Initial startup

Energy storage power station: Activated via BMS, pre-charging on the DC side, then the grid-connected inverter starts up.

Establish voltage and frequency: The initial voltage is set to 95% of the rated value, and the frequency is stabilized at 50±0.5 Hz.

· Load and network reconfiguration

Critical load restoration: Gradually restore loads according to priority (each step increases load by no more than 10% of the power source capacity).

Substation charging: Power is gradually supplied to the substation from the black start power source in sequence (low voltage → medium voltage → high voltage). Use the zero-rise voltage regulation strategy to avoid transformer inrush current 冲击 (e.g., limit the voltage rise rate to ≤5%/s).

Generator start-up: Cold reserve power generation units require plant power supply provided by black start power sources.

Synchronization check: Newly started units must meet the synchronization conditions of voltage, frequency, and phase angle difference less than 10° before being connected to the grid.

· Expansion and grid integration

Zone Expansion: Expand the power supply area using either "serial recovery" or "parallel recovery" strategies.

Main grid connection: Synchronize and connect to the main grid via tie lines when the local grid is stable and has sufficient capacity.

10.4 Black Start - EMS Operating Procedures

(1) Synchronous start-up

To minimize timing differences in wave generation among different machines, during synchronization, all machines must first be brought into standby mode. Only after confirming that all machines have entered standby mode should the start command be issued.

For applications involving larger transformer capacities, multiple PCS units need to start simultaneously to support the inrush current during the transformer magnetization process. The EMS system sends start commands synchronously to all PCS units, ensuring that the maximum communication delay between any two PCS units is less than 100 ms.

(2) Asynchronous start-up

PCS are started asynchronously, one by one. The EMS sends start commands to the PCS individually. After one unit completes startup, the next unit starts, and so on, until all units have been started.

This operating condition applies when the black-start capacity is relatively small, and a single unit of PCS is sufficient to accomplish the black-start of the entire station's equipment or load.

11. Islanding / Anti-islanding / Droop mode

● Anti-islanding Overview

Anti-islanding is a core protection function of grid-connected power electronic devices (such as photovoltaic inverters and energy storage PCS), with the primary goal of preventing distributed power sources from continuing to supply local loads after grid power outage, thus avoiding the formation of an uncontrolled "island."

● Islanding risk

Definition: When the power grid is disconnected due to a fault or planned outage, distributed generators (such as photovoltaics and energy storage) continue to supply power to local loads, forming an isolated "island" operation.

Hazard:

Personal safety: Maintenance personnel might incorrectly assume the line is de-energized, leading to the risk of electric shock.

Equipment damage: Islanding operation may cause voltage/frequency instability, damaging loads or power equipment.

Reclosing surge: Current surge occurs when the grid reconnects unsynchronized with the island.

● **Anti-islanding implementation principle**

Core logic: Real-time monitoring of grid status, rapidly disconnecting the grid-tie switch upon detecting a power outage (typically requiring an operation time ≤ 2 seconds).

Detection method:

Passive detection: Monitoring abnormalities in grid parameters (voltage, frequency, harmonics). Over/under voltage protection, over/under frequency protection, voltage phase sudden change detection;

Active detection: Inject small disturbances into the grid and observe the system's response. Active Frequency Disturbance (AFD) Sliding Mode Frequency Shift (SMS), current harmonic injection.

● **IES900 Anti-islanding Basic Status**

Our company's IES900 provides a wide range of adjustable parameters, meeting the requirements for anti-islanding protection under various scenarios, in accordance with regulatory requirements and actual on-site conditions. The parameter settings are shown in Figure 4, Anti-Islanding Parameters.

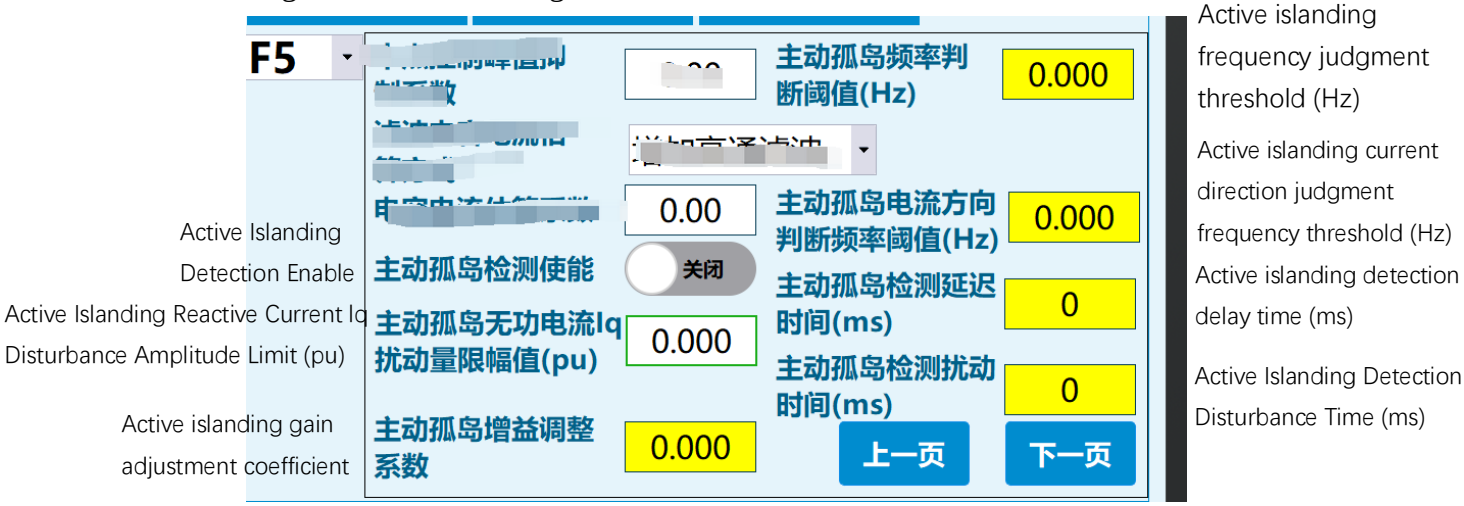


Figure 4 Anti-islanding Effect Parameters

11. VSG Function

11.1 Overview of Virtual Synchronous Machines

The Virtual Synchronous Generator (VSG) function of a PCS (Power Conversion System) is a technology that simulates the operating characteristics of a synchronous generator through control algorithms, aiming to enhance the stability of power systems, especially in scenarios involving renewable energy integration.

11.1.1 Basic Principles

Emulating synchronous generator characteristics: By means of control algorithms, the PCS is enabled with inertia, damping, and primary frequency and voltage regulation capabilities similar to those of synchronous generators, thus providing dynamic support during power grid fluctuations.

Core objective: Address frequency/voltage stability issues caused by the lack of inertia when integrating new energy sources (such as photovoltaics and wind power) into the grid.

11.1.2 Parameters and Performance Impact

Virtual inertia (J): The larger the value, the stronger the inertia response, but the dynamic response speed may become slower.

Damping coefficient (D): The larger the value, the stronger the oscillation suppression capability, but it may reduce response sensitivity.

Tuning coefficient: Determines the slope of active/reactive power variation with frequency/voltage.

11.1.3 Typical Application Scenarios

New energy power plants: Provide inertial support for photovoltaic and wind power, improving grid integration stability.

Microgrid: Maintains frequency/voltage during islanded operation, enhancing self-sufficiency.

Black start: Simulates synchronous machine characteristics to assist in restoring power to the grid.

Energy storage system: Responds rapidly to grid fluctuations through quick charging and discharging, optimized for regulation using the VSG algorithm

11.2 Detailed information on virtual synchronous generators

Table 1 Comparison between Grid-Forming and Grid-Supporting Converters

Grid-following converter	Grid-forming converter
The external characteristic of a grid-following converter is a controlled current source.	The external characteristic of a grid-forming converter is a controlled voltage source.
Unable to provide voltage support and frequency support to the power grid	Can provide voltage support and frequency support to the power grid.
Under weak grid conditions, larger line impedance degrades the stability of the current loop.	Under weak grid conditions, line impedance has a minor impact on the voltage loop, allowing the voltage loop to maintain good stability.
Grid-following converters achieve synchronization with the grid through a phase-locked loop (PLL).	Grid-forming converters achieve synchronization with the grid using power synchronization methods.

Table 1 compares grid-following converters and grid-forming converters. The external characteristic of a grid-forming converter is a controlled voltage source. When loads are connected to or disconnected from the grid, the grid-forming converter, acting as a voltage source, can rapidly respond to load changes, thereby reducing the burden on synchronous generators and decreasing the fluctuation amplitude of grid frequency and voltage. Currently, the control strategies for grid-forming energy storage converters mainly include droop control and virtual synchronous machine (VSM) control. Since these two control methods are mathematically equivalent, our company's grid-forming energy storage converters adopt the virtual synchronous machine control.

Voltage-controlled virtual synchronous generators can be classified into two types: those without inner loops and those with inner loops. The schematic diagram of a virtual synchronous machine without an inner current loop is shown in Figure 5. This control structure is the simplest and easiest to implement, but it lacks a current inner loop, resulting in no overcurrent protection mechanism at the control level.

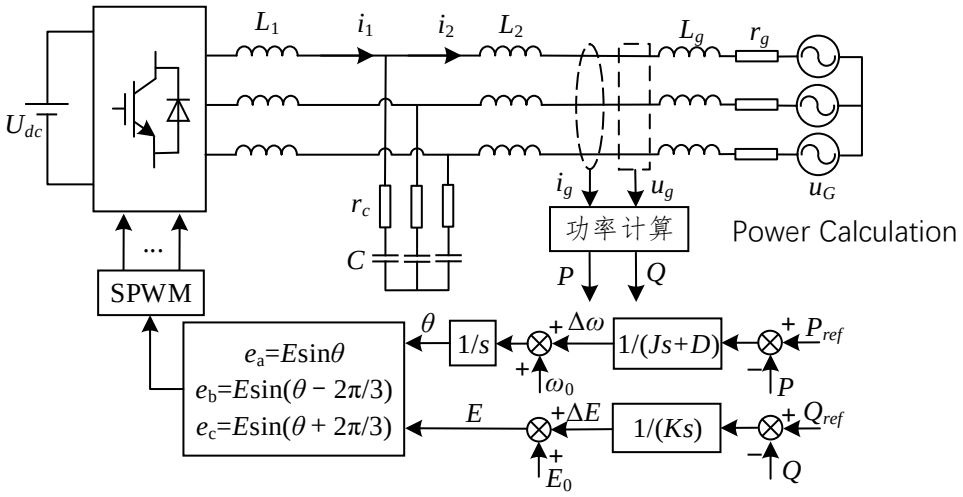


Figure 5: Virtual Synchronous Machine without Inner Loop

The schematic diagram of the virtual synchronous machine with an inner loop is shown in Figure 6. In this control structure, the virtual synchronous machine control serves as the outer loop, while the capacitor voltage control and machine-side current control function as the inner loops. In this control structure, the capacitor voltage is equivalent to the electromotive force of a synchronous generator, and the line impedance along with the grid-side inductance of the filter is equivalent to the impedance of the synchronous generator. During transient processes, when loads are switched, voltage-controlled virtual synchronous machines can rapidly respond to sudden increases or decreases in load, providing both active and reactive power support to the grid.

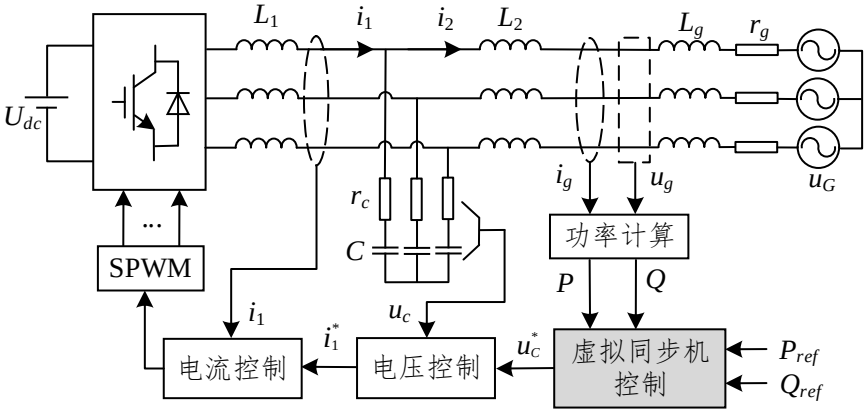


Figure 6 Voltage-controlled virtual synchronous machine

Currently, voltage-controlled virtual synchronous machines are the mainstream solution in grid-forming energy storage converters, so our company has adopted this control scheme. The control strategy for a battery energy storage system based on grid-

forming converters is shown in the figure below, and the control strategy includes the following steps:

1. Active Power Control Section

a) Virtual FM control section. Simulate the governor characteristics of a synchronous generating unit's prime mover, simplifying consideration of its active power–frequency characteristics. The converter detects the actual grid frequency, compares it with the reference frequency, passes through a deadband control module, and if the frequency deviation exceeds the set deadband width, it outputs the actual deviation; otherwise, it outputs zero. Multiplying the output frequency deviation by the droop coefficient yields the additional power reference value. The energy storage converter achieves primary frequency regulation by autonomously responding to changes in the power system frequency. The virtual FM control model is expressed as follows:

$$\Delta P_{ref} = K_p(f^* - f) | f^* - f | \geq f_{deadzone}$$

Where ΔPref is the additional reference power, f* is the reference frequency, f is the actual frequency calculated from the voltage signal acquired from the grid side, fdeadzone is the set frequency deadband; Kp is the droop coefficient.

b) Virtual inertia and damping control section. Simulate the rotor motion equation of a synchronous generator; introduce inertia and damping components in control to enhance the flexible grid-connected characteristics of the converter. The sum of the additional reference power obtained from the virtual FM control loop and the active power reference command is taken as the converter's virtual mechanical power, while the actual active power output of the converter is taken as the virtual electromagnetic power. The difference between them, after passing through the inertia and damping control loops, yields the virtual speed of the energy storage converter, and integrating this gives the virtual internal electromotive force phase angle. The control model is expressed as follows:

$$J\frac{d\omega}{dt} = P_m - P - D_{Equ}(\omega_0 - \omega)$$

$$\frac{d\theta}{dt} = \omega$$

$$P_m = P_{ref} + \Delta P_{ref} = P_{ref} + K_p(f^* - f)$$

Here, J represents the virtual inertia, ω denotes the virtual angular velocity of the converter's output internal electromotive force; Pm is the input mechanical power corresponding to a conventional synchronous machine; P is the actual active power output of the converter; DEqu is the equivalent virtual damping coefficient; ω0 is the rated angular velocity of the system; Θ is the virtual phase angle of the internal electromotive force output from the energy storage converter; Pref is the active power reference command; ΔPref is the additional reference power; Kp is the droop coefficient; f(*) is the reference frequency; f is the measured frequency.

To enhance the virtual damping control capability, Equ D consists of two parallel parts:

$$D_{Equ} = D + \frac{T_w s}{1 + T_w s} \cdot \frac{K_D(1 + T_3 s)}{1 + T_4 s}$$

Here, D is the virtual damping coefficient, Tw is the time constant of the DC-blocking section, T3 and T4 are the time constants of the phase-shifting sections, and KD is the speed deviation amplification factor.

2. Reactive Power Control Section

Simulate the characteristics of a synchronous generator excitation system, and introduce the electromagnetic equations from the three-order practical model of the synchronous generator to accurately simulate excitation dynamic characteristics:

$$T'_{d0} \frac{dE'_q}{dt} = E_{qe} - E'_q - i_d(x_d - x'_d)$$

Here, T'd0 is the excitation winding time constant; E'q is the transient voltage; Eqe is the forced no-load electromotive force linearly related to the excitation voltage; id is the d-axis current component; xd is the synchronous reactance; x'd is the d-axis transient reactance.

First, the voltage and current at the output terminal of the energy storage converter are collected. After passing through the virtual impedance droop adjustment stage, the actual voltage signal of the virtual excitation regulator is obtained: the voltage deviation caused by the virtual impedance is locally compensated by the energy storage converter; the frequency and voltage deviations caused by the droop characteristics are uniformly compensated by the secondary frequency and voltage regulation of the upper-level power coordination control system. The virtual impedance is compensated using a high-

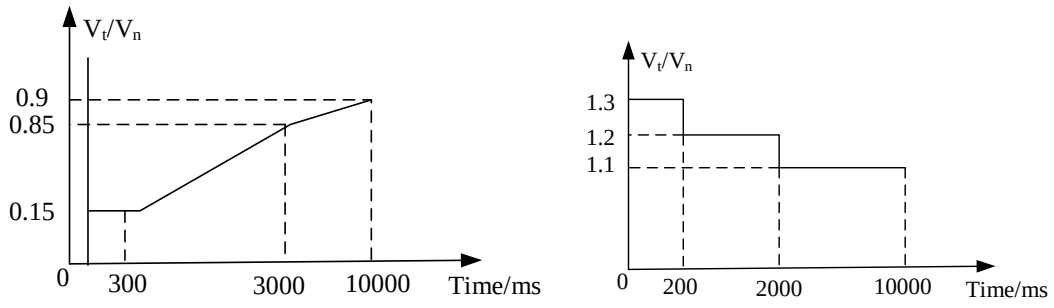
pass filter method, which only corrects the voltage deviation caused by virtual impedance under steady-state conditions. By properly designing the filter type and parameters, the decoupling capability during transients and the ability to suppress parallel circulating currents can be preserved during compensation.

In response to market demand for grid-forming energy storage converters, our company has initiated scientific research projects to conduct systematic technical studies on grid-forming energy storage converters. Various functions have been verified through hardware-in-the-loop simulation and physical experiments, confirming compliance with national standards.

12. LVRT/HVRT/FVRT/ZVRT

12.1 Fault Ride-Through Curve

According to the CEA 2022 standard, power generation equipment must have the capability to ride through high and low voltage conditions. As shown in the figure below, for both high and low voltage ride-through requirements, when the grid connection point voltage is above the limit line, the power station must remain connected to the grid and must comply with the restrictions indicated by the dashed line in the figure. Reactive power support is required to maintain voltage during grid faults or sudden changes.



a) CEA Low Voltage Ride-Through Curve

b) CEA High Voltage Ride-Through Curve

Figure 7 CEA fault ride-through curve

12.2 Reactive Current Support During Crossing

Symmetrical fault	Active and reactive power control strategies during low-voltage ride-through	Dynamic reactive current increment △It = K1*(1.0 - Ut)*In, when Ut < 0.6, dynamic active current It = 0.1*In When Ut > 0.6, the dynamic active current should be fully dispatched according to the maximum apparent power capability.
	Active and reactive power control strategies during high penetration	Dynamic reactive current increment △It = K3 * (Ut - 1.0) * In Dynamic active current I = ±Imax * In (|original active current| > Imax) = original active current (|original active current| < Imax)
Asymmetrical fault	Active and reactive power control strategies during low-voltage ride-through	Dynamic positive-sequence reactive current increment △It = K1*(1 - Ut_pos)*In Dynamic negative sequence reactive current increment △It = K2 * Ut_neg * In When Ut < 0.6, the dynamic active current It = 0.1 × In When Ut > 0.6, the dynamic active current should be fully dispatched according to the maximum apparent power capability.
	Active and reactive power control strategies during high penetration	Dynamic reactive current increment △It = K3 * (Ut - 1.0) * In Dynamic negative sequence reactive current increment △It = K2 * Ut_neg * In Dynamic active current It = ±Imax × In (|original active current| > Imax) = original active current (|original active current| < Imax)

Imax is the active current limit value for fault ride-through. Function code is configurable
The parameter K1 can be set as the positive-sequence reactive current injection coefficient during low voltage ride-through. Function code is configurable
K2 can be set as the negative-sequence reactive current injection coefficient during low voltage ride-through. Function code is configurable
K3 can be set as the reactive current injection coefficient during high voltage ride-through. Function code is configurable
The parameter Ut represents the per-unit voltage at the connection point of the energy storage converter
. Ut_pos represents the per-unit positive-sequence voltage at the connection point of the energy storage converter.
Ut_neg represents the per-unit negative-sequence voltage at the connection point of the energy storage converter.